RAKESH CHOWDARY POLINA

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EDUCATION

Master of Science in Computer Science **May 2024**
Oklahoma State University, Stillwater, OK **CGPA: 3.909/4**

SKILLS

Languages:	Python, Java, JavaScript, SQL, C, C++, HTML, CSS, Swift
Frameworks:	Flask, Django, React JS, Angular JS, Node JS, Spring boot, FastAPI, React Native
Databases:	MySQL, MongoDB, NoSQL, PostgreSQL, Cassandra
Web Technologies:	RESTful API, Django ORM, SQLAlchemy, OAuth 2, jQuery, JMS, RabbitMQ
Web Services:	REST, SOAP
Tools/Platforms:	Docker, Git, JIRA, Jenkins, Postman, Ansible, Kubernetes, Apache Kafka, Terraform, Swagger
AWS Cloud Services:	EC2, S3, EKS, RDS, ELB, Cloud Watch, CloudFront
Build Tools:	Pip, Poetry

WORK EXPERIENCE

Research Data Analyst, Oklahoma State University **May 2023 – May 2024**

- Engineered and implemented a robust facial recognition pipeline, performing both quantitative and qualitative analysis on image data with Python and InsightFace, to achieve a precision rate of 93.6%
- Implemented DBSCAN clustering and rigorously verified data accuracy across 182,000+ images, enhancing archive reliability and insight precision.
- Designed an intuitive UI for the labeling tool using Flask and React.js, streamlining user engagement and annotations.
- Designed and maintained database structures using SQL, enhancing user interaction management and feedback data integrity, ensuring operational excellence.

Software Engineer, Ericsson, India **Jul 2019 – Jul 2022**

- Assisted in migrating legacy monolithic systems to modular architectures using Python and Flask, designing RESTful APIs for critical internal tools. Integrated RabbitMQ for asynchronous task handling, improving system scalability to support 30% more concurrent users.
- Collaborated with UX teams to implement front-end enhancements for customer-facing dashboards using React and TypeScript. Authored user guides and conducted training sessions for internal stakeholders to streamline adoption of new features.
- Analyzed business requirements and translated them into technical specifications for API and database solutions. Optimized PostgreSQL query performance by 25% and reduced downtime through collaboration with vendor support teams during database version upgrades.
- Developed and maintained secure, scalable APIs using Django and Flask, incorporating authentication/authorization via OAuth2. Reduced API latency by 25% through connection pooling and caching strategies. Created API documentation using Swagger for developer teams.
- Containerized applications using Docker and managed deployments via AWS ECS. Partnered with DevOps to troubleshoot production issues, reducing deployment failures by 20%.
- Automated CI/CD pipelines using Jenkins and GitLab CI, integrating unit/integration testing with pytest. Contributed to test plan creation for major releases, documenting 50+ test cases for regression testing.
- Provided Tier 2/3 technical support for enterprise web applications, resolving 150+ tickets related to system integrations, data discrepancies, and authentication issues. Automated routine tasks using Python and PowerShell scripts.
- Enforced security protocols for user access controls and data encryption, passing 100% of internal compliance audits. Remediated vulnerabilities identified during penetration testing.
- Managed sprint deliverables as SCRUM lead for 3 cross-functional teams, coordinating with product owners to prioritize bug fixes and feature requests in Jira.
- Developed system requirements for migration of legacy on-prem systems to AWS, including disaster recovery workflows. Created technical runbooks and trained operations teams on monitoring tools like Grafana and CloudWatch.

PROJECTS

Sleep data analysis - Applied LSTM network and Granger causality techniques to investigate sleep data on a dataset comprising over 5,000 individuals' sleep patterns, unveiling insights into temporal dynamics and causal connections within the dataset

HireOSUgrads App - Developed an iOS application for university students and recruiters, featuring intuitive swipe gestures for seamless interactions. Utilized Swift, followed MVC architecture and integrated SQLite for robust data management

Journal App - Developed an android application, that allows users to store their daily events and memories using React Native, Express JS, Redux, Redis, MongoDB, REST APIs, Git.